

Interface Capability Brief

EDI 835 Payment Integrity · Stakeholder overview · 2026-08-13

1. Executive summary

Ingest ERA 835s, enrich each claim payment with open-AR lineage from SQL Server, bucket matched vs exception, and emit an underpay alert CSV for shortfalls finance can act on. Demo only — not a full posting engine.

2. Who this is for

- Sources: Remit drop folder (input/inbound/*.edi) - Destinations: output/matched/, output/exceptions/, output/underpay/underpay_alerts.csv, OpenAR status updates - Demo vs production: Demo only — synthetic 835 + AR, no PM/EHR posting, no SNIP Types 1–7 for 835

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3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: Directory / File (DirectoryListener)
- Format / envelope: X12 835 (5010), multi-ST //Transaction fork
- Identity fields: CLP02 patient control # → OpenAR.ClaimControlNumber
- Samples path: samples/sample_835_underpays.edi

Integration highlights

FQCN sources: edi-276-277-claim-status, csv-sftp-to-sql, PilotFish Documentation 26R1.11. Image: pilotfish-eip:23R1.

Stage	Module / mechanism	Notes
R1 Listener	com.pilotfish.eip.modules.file.DirectoryListener	Poll \$EDI_INBOUND_DIRECTORY. Skeleton: edi-276-277-claim-status
R1 Format	EDITransformationModule + XPathForkingModule //Transaction	TableData edi-tabledata/835-W1 (5010). Playbook §3.6. Docs 26R1.11 vs image 23R1.
R1 Target processors	XPath → DatabaseSqlProcessor OpenAR lookup → XSLT score	Build RemitDecision. SQL processor skeleton: csv-sftp-to-sql JDBC tags
R1 Transport	DirectoryTransport + XML Formatting on the transport	Stage under output/staged-decisions/ (playbook §1.4)
R2 Listener	DirectoryListener	Poll staged decisions
R2 Router	XPathRoutingModule accumulate=true	matched / exception / underpay CSV
R2 Transports	DirectoryTransport + NullTransport + SQL UPDATE	XML buckets get XML Formatting; CSV does not

4. Honest boundaries (not claimed yet)

- Add selective mock PM post in a later pass?

- Fork //CLP (+ sibling CAS context) for true multi-claim ST files?

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - Poll And Score Remits” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- What is checked: EDI parses; CLP identity present; AR presence; paid vs expected within \$0.01 - What is NOT checked: SNIP 835, PLB full accounting, CAS semantics beyond string capture, dual-write posting to PM - Does failure block outbound? yes for unparseable EDI; no (exception theater) for business exceptions

6. State & idempotency

- Status model: OpenAR.Status OPEN → MATCHED | UNDERPAY | EXCEPTION - When state advances: after RemitDecision is built, on each routing target - Dedup keys: ClaimControlNumber + inbound file stamp (resubmits re-append CSV / overwrite decision files) - Retry / poison: bad EDI fails at EDI→XML (fail-closed); unmatched AR stays EXCEPTION

8. Dual-write / side effects

- Order: decision XML → route fan-out (files + SQL update) - Compensation: none (demo) - Demo shortcuts: underpay threshold fixed at \$0.01; CSV append without unique transaction id enforced

3. Inbound contract

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Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - Poll And Score Remits: runs steps such as Poll EDI 835 Files, Fork - XPath, Stage Remit Decision.

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2 - Route Remit Decisions: runs steps such as Poll Staged Remit Decisions, Conditional Router, Write Matched, Update AR Matched.

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7. Security & trust posture

- X12 SNIP validation is used where configured to reject bad EDI before downstream work
- Credentials and passwords in this package are demo-only — not production secrets
- Risk noted in design: High — No PM/EHR post — Story stops at buckets + CSV
- Risk noted in design: Med — 23R1 trial X12 tables expired — Sandbox mounts EDI/TableData/x12 → eip-root/edi-tabledata
- Risk noted in design: Med — Claim-before-complete on AR UPDATE — File write and SQL update not XA
- Risk noted in design: Med — Multi-CLP per ST not forked per CLP — Fork is //Transaction
- Risk noted in design: Low — CSV append races — Concurrent ST

8. Seeing it work

- - Ports: SQL 14339, EIP 8110, Web UI 8111 - Volumes: ./input, ./output, ./logs, ./samples, ./documents, TableData mount - Heap: 512M–2GB - Dependencies / cold start: SQL health + seed ~30–60s, EIP ~60–90s
- Local: http://127.0.0.1:8111/
- EIP: http://127.0.0.1:8110/eip/
- SQL: 127.0.0.1:14339 (sa / PilotFish_Demo1! · Edi835PaymentIntegrity)
- docker compose --profile full up -d --build

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.